

SP-04: A generic counterexample to the smallest-absolute-multiplier rule

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Status of this manuscript: Independently checked resolution (Codex-agent review).

Outcome: Negative resolution.

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The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently checked the complete proof against the exact repository target on 11 September 2026; its [detailed review](#) records **PASS**. The theorem and proof text below are unchanged from the reviewed submission. This verification is an independent agent review, not external human peer review or a formal proof certificate.

Theorem SP-04: proposed generic counterexample

For real data U , consider real stationary pairs

$$|\det X| = 1, \quad X^T(U - X) = cI.$$

The rule under consideration selects a pair with smallest $|c|$ and claims that its matrix is nearest to U in Frobenius norm.

Claim. The rule fails for every diagonal matrix

$$U = \text{diag}(s_1, s_2, s_3), \quad \frac{7}{4} < s_1 < s_2 < s_3 < \frac{44}{25}.$$

It therefore fails on a nonempty open set of real 3×3 matrices and cannot hold outside a proper real algebraic exceptional set.

1. Reduction of all stationary matrices

Write $S = X^T X$. The stationary equation gives

$$U = X + cX^{-T}, \quad U^T U = S + 2cI + c^2 S^{-1}.$$

Thus S commutes with $U^T U$. The latter is diagonal with distinct entries, so S is diagonal. Also

$$U = X(I + cS^{-1}).$$

The second factor is invertible, because both U and X are invertible. Hence X is diagonal as well. Writing its entries as x_i , every stationary pair satisfies

$$x_i^2 - s_i x_i + c = 0, \quad |x_1 x_2 x_3| = 1.$$

2. No stationary pair has $0 \leq c \leq 13/25$

If $c = 0$, invertibility forces $X = U$, whose determinant exceeds one. For $0 < c \leq 13/25 = 0.52$, write

$$r_i(c) = \frac{s_i + \sqrt{s_i^2 - 4c}}{2}.$$

The two roots are r_i and c/r_i , both positive. Throughout this range, $r_{\min} > 1$ and $r_{\max} \leq 1.76$. For example,

$$r_{\min} > \frac{1.75 + \sqrt{1.75^2 - 4(0.52)}}{2} > 1.$$

Moreover,

$$\frac{\partial r}{\partial s} = \frac{1}{2} \left(1 + \frac{s}{\sqrt{s^2 - 4c}} \right) < 2,$$

since $s^2 - 4c \geq 0.9825$ and $s \leq 1.76$. The interval of possible s_i has width 0.01, so

$$r_{\max} - r_{\min} < 0.02, \quad \frac{r_{\max}}{r_{\min}} < 1.02.$$

Choosing three large roots gives product greater than one. Any selection containing a small root has product at most the largest product with exactly one small root, since replacing a small root by its large partner increases the product. That product is bounded by

$$c \frac{r_{\max}^2}{r_{\min}} < 0.52(1.76)(1.02) = 0.933504 < 1.$$

No root selection has product one.

3. The unique least absolute multiplier is negative

Set $c = -t$, $t > 0$, and put

$$a_i(t) = \frac{s_i + \sqrt{s_i^2 + 4t}}{2}.$$

The two roots are $a_i(t)$ and $-t/a_i(t)$. Choose the negative root only at the first index. The absolute determinant is

$$g(t) = \frac{t}{a_1(t)} a_2(t) a_3(t).$$

Every factor is strictly increasing for $t > 0$: in particular,

$$\frac{t}{a_i(t)} = \frac{\sqrt{s_i^2 + 4t} - s_i}{2}.$$

The continuous function g is strictly increasing from zero. At $t = 0.52$, the positive roots exceed two, because $4 - 2s_i - 0.52 < 0$. The magnitude of the negative root exceeds $1/4$, since

$$(1/4)^2 + s_1/4 < \frac{1}{16} + \frac{1.76}{4} = 0.5025 < 0.52.$$

Consequently $g(0.52) > 1$, and there is a unique $t_* \in (0, 0.52)$ with $g(t_*) = 1$.

For a nonempty set E of negative-root indices, the absolute determinant at fixed $t > 0$ is

$$a_1 a_2 a_3 \prod_{i \in E} \frac{t}{a_i^2}.$$

Each extra factor is strictly between zero and one, and the largest factor is uniquely t/a_1^2 , since $a_1 < a_2 < a_3$. Thus, among all nonempty negative-root patterns, the absolute determinant is uniquely largest for $E = \{1\}$. Each pattern's product increases strictly with t and tends to infinity. All other patterns therefore reach one strictly after t_* . The all-positive pattern has product greater than one for every $t > 0$ and gives no such stationary pair.

Together with the exclusion of $0 \leq c \leq 0.52$, this proves that $c_* = -t_*$ is the unique smallest-absolute stationary multiplier.

4. Failure of nearestness and the generic qualifier

At this selected pair, $x_1 < 0$ and the other two entries are positive. Replacing x_1 by $|x_1|$ preserves $|\det X| = 1$ and reduces the squared distance by

$$(s_1 - x_1)^2 - (s_1 - |x_1|)^2 = 4s_1|x_1| > 0.$$

The selected matrix is therefore not nearest.

For the concrete rational data

$$(s_1, s_2, s_3) = (1751, 1755, 1759)/1000,$$

the diagnostic numerical values are

$$\begin{aligned} c_* &\approx -0.497407247776, \\ (x_1, x_2, x_3) &\approx (-0.24873641, 2.00329461, 2.00685420), \\ \|U - X_*\|_F^2 &\approx 4.122027613290. \end{aligned}$$

The feasible matrix I_3 already has smaller squared distance 1.710107. These decimals illustrate the exact inequalities above; they are not used as proof.

Finally, if $U = P \operatorname{diag}(s) Q^T$ is a singular-value decomposition, the transformation $Y = P^T X Q$ preserves $|\det X|$, the Frobenius objective, and the stationary multiplier:

$$Y^T (\operatorname{diag}(s) - Y) = Q^T X^T (U - X) Q = cI.$$

The same failure occurs for every real matrix with distinct singular values in $(7/4, 44/25)$. This set is nonempty and open. A proper real algebraic set has empty Euclidean interior, so algebraic genericity cannot exclude all these counterexamples. $\square$

Scope and review notes

Both determinant signs are included, as required by the entry. The proof gives a unique least absolute multiplier and extends from diagonal data to an open set of matrices with distinct singular values. This addresses the stated algebraic-generic qualifier rather than only an exceptional example. No claim is made about a different projection problem restricted to determinant +1.

The independent review checked the reduction of every stationary matrix (not just a diagonal candidate), uniqueness of the smallest absolute multiplier, and the passage to an open set under left/right orthogonal transformations.

The packaging pass checked the repository statement and contribution rules on 11 September 2026. It did not conduct a new exhaustive literature search or establish novelty.

Source references

Alex Townsend, *Open Problems in Numerical Linear Algebra* (2026), [entry SP-04](#), repository snapshot b412319, accessed 11 September 2026.

J. A. Baaijens and J. Draisma, *Euclidean distance degrees of real algebraic groups*, *Linear Algebra and its Applications* 467 (2015), 174–187, §4.1 and Problem 4.3 on p. 186, [DOI](#). The catalog also cites P. Jaap and O. Sander, *How to project onto $SL(n)$* , [arXiv:2501.19310](#), introduction, for the related determinant-one problem. Source locators are those recorded by the catalog.